Burga Yamini Lakshmi

7013957666 | Vijayawada, Andhra Pradesh

yaminiburga@gmail.com | www.linkedin.com/in/burga-yamini-lakshmi-9432a1291

EDUCATION

VIT-AP UNIVERSITY

BTECH IN ECE(VLSI) | VIT-AP
CGPA:8.32/10 | (2023 - 2027) | Amaravati

SRI CHAITANYA COLLEGE

CLASS XII | MPC
Percentage:93.8 | 2023 | Guntur

ZPHS SCHOOL

CLASS X
CGPA:10/10 | 2023 | Pandillapalli

SKILLS

TECHNICAL SKILLS:

Programming Languages:

- C
- C++
- Python
- Java

HDL Languages:

- Verilog HDL
- SystemVerilog (Basic)

Digital Design:

- RTL Design
- FSM Design
- FPGA Fundamentals
- Digital Logic Design

VLSI:

- CMOS VLSI
- Semiconductor Fundamentals
- AMBA Bus Protocols (AHB/APB)

Tools:

- Xilinx Vivado
- ModelSim

SOFT SKILLS:

- Problem-Solving
- Collaboration
- Communication

CERTIFICATION

- RTL Design & Logic Verification Internship – Maven Silicon.
- Drone Technology Internship – Indian Space Lab.
- AWS Academy Graduate – AWS Academy Cloud Foundations (Amazon Web Services).

PROJECTS

5-Stage Pipelined Processor using Verilog HDL

- Designed and implemented a 32-bit 5-stage pipelined RISC processor using Verilog HDL with IF, ID, EX, MEM, and WB stages.
- Implemented hazard detection, data forwarding, and pipeline registers, and verified the 5-stage pipelined processor on an FPGA board using Vivado.

AHB to APB Bridge using Verilog HDL

- Designed an RTL-based AHB–APB Bridge following the AMBA protocol using Verilog HDL.
- Implemented FSM-based control logic and verified protocol compliance through simulation.

Aircraft Detection & Radar System using Arduino

- Developed an Arduino-based radar system with 180° scanning using ultrasonic sensors and a servo motor.
- Built a Python GUI for real-time radar visualization through serial communication.

Smart Vertical Farming

- Built an Arduino Uno-based hydroponics farming system that grows plants without soil by supplying nutrient-rich water, while using sensors to monitor growing conditions.
- Integrated water pumps, LED grow lights, and relay modules to automate water circulation, nutrient supply, and lighting, allowing plants to grow efficiently without direct sunlight.

POSITION OF RESPONSIBILITY

- Core Team Member (DSA CLUB VIT-AP) **(2024 - 2025)**